

Next Generation Data Centers and Energy Management Systems

Lecture 2B: Psychrometrics and Heat Transfer

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- **Psychrometrics and Heat Transfer**
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Analyze a data center CRAC and rack using thermodynamic and psychrometric principles.
- Analyze the cooling system within the data center hall using a thermofluid flow network.

Water-side economizer

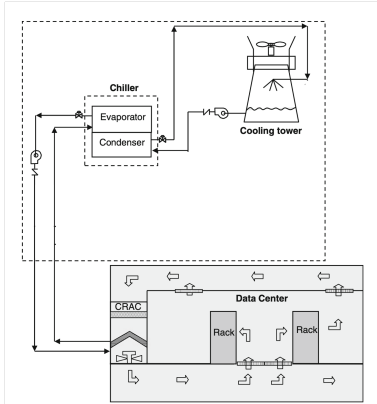


Figure 1: No economizer (Figure 2.53 in Ref. [1]).

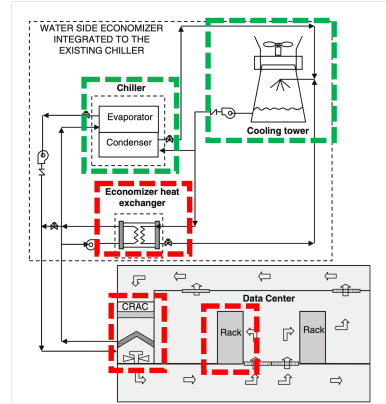


Figure 2: Water-economizer (Figure 2.53 in Ref. [1]).

Energy analysis of a CRAC

Energy analysis of a rack

Flow network theory

Flow network analysis of a data center hall

References I

- [1] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.